

Launch AWS Acad

Properties > spar

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practicalexam - D

Databricks - Your

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dbc-6784a36d-47b3.cloud.databricks.com/explore/data/volumes/pratical/schema1/practicalexam?o=7474647174911736

workspace

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Spaces

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

Catalog

Serverless Starter Warehouse Serverless 2XS

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For you All

My organization

workspace

system

ctlg1

newcatalog

practical

default

information_schema

schema1

practicalexam

Delta Shares Received

samples

Catalog Explorer > practical > schema1 >

practicalexam

Overview Files Details Permissions

Description

AI generate Add

/Volumes/pratical/schema1/practicalexam

Refresh Create directory

Filter files and directories at this level

Name	Size	Last modified
airports.csv	12.03 MB	32 seconds ago
runways.csv	3.76 MB	33 seconds ago

About this volume

Owner shraddha256@gmail.com

Tags

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ENG IN 11:27 09-05-2026

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- Home
- Learn
- Workspace
- Recents
- Catalog
- Jobs & Pipelines
- Compute
- Discover Beta
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie Spaces
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Runs
- Data Ingestion

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Run all Serverless Schedule Share

11:45 AM (1s) 1

```
from pyspark.sql.functions import *
from pyspark.sql.types import *
```

11:45 AM (39s) 2

```
df_airports = spark.read.csv(
    "/Volumes/practical/schemal/practicalexam/airports.csv",
    header=True,
    inferSchema=True
)
```

> df_airports: pyspark.sql.connect.dataframe.DataFrame - [id: integer, ident: string ... 17 more fields]

11:46 AM (2s) 3

```
df_runways = spark.read.csv(
    "/Volumes/practical/schemal/practicalexam/runways.csv",
    header=True,
    inferSchema=True
)
```

> df_runways: pyspark.sql.connect.dataframe.DataFrame - [id: integer, airport_ref: integer ... 18 more fields]

11:46 AM (1s) 4

```
df_airports = df_airports \
```

Hot weather Now

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ENG IN 12:03 09-05-2026

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- Learn
- Workspace
- Recents
- Catalog
- Jobs & Pipelines
- Compute
- Discover Beta
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie Spaces
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Runs
- Data Ingestion

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Run all Serverless Schedule Share

11:46 AM (1s) 4

```
df_airports = df_airports \
    .withColumn("latitude_deg", col("latitude_deg").cast("double")) \
    .withColumn("longitude_deg", col("longitude_deg").cast("double")) \
    .withColumn("elevation_ft", col("elevation_ft").cast("integer"))

df_runways = df_runways \
    .withColumn("length_ft", col("length_ft").cast("integer")) \
    .withColumn("width_ft", col("width_ft").cast("integer"))
```

> df_airports: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 17 more fields]
> df_runways: pyspark.sql.connect.dataframe.DataFrame = [id: integer, airport_ref: integer ... 18 more fields]

11:46 AM (<1s) 5

```
print("AIRPORTS SCHEMA")
df_airports.printSchema()
```

root

```
-- id: integer (nullable = true)
-- ident: string (nullable = true)
-- type: string (nullable = true)
-- name: string (nullable = true)
-- latitude_deg: double (nullable = true)
-- longitude_deg: double (nullable = true)
-- elevation_ft: integer (nullable = true)
-- continent: string (nullable = true)
-- iso_country: string (nullable = true)
-- iso_region: string (nullable = true)
-- municipality: string (nullable = true)
```

12:04 09-05-2026

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- Workspace
- Recents
- Catalog
- Jobs & Pipelines
- Compute
- Discover Beta
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie Spaces
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Runs
- Data Ingestion

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```
-- gps_code: string (nullable = true)
-- local_code: string (nullable = true)
-- home_link: string (nullable = true)
-- wikipedia_link: string (nullable = true)
-- keywords: string (nullable = true)
```

Run all Serverless Schedule Share

11:48 AM (-1s) 6

```
print("RUNWAYS SCHEMA")
df_runways.printSchema()
```

```
-- id: integer (nullable = true)
-- airport_ref: integer (nullable = true)
-- airport_id: string (nullable = true)
-- length_ft: integer (nullable = true)
-- width_ft: integer (nullable = true)
-- surface: string (nullable = true)
-- lighted: integer (nullable = true)
-- closed: integer (nullable = true)
-- le_id: string (nullable = true)
-- le_latitude_deg: double (nullable = true)
-- le_longitude_deg: double (nullable = true)
-- le_elevation_ft: integer (nullable = true)
-- le_heading_degI: double (nullable = true)
-- le_displaced_threshold_ft: integer (nullable = true)
-- he_id: string (nullable = true)
-- he_latitude_deg: double (nullable = true)
-- he_longitude_deg: double (nullable = true)
-- he_elevation_ft: integer (nullable = true)
-- he_heading_degI: double (nullable = true)
-- he_displaced_threshold_ft: integer (nullable = true)
```

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Big Data Tech

Crimes - 202

Databricks - Y

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New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Spaces

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

Hot weather Now

practical lab

File Edit View Run Help Python

Last edit was now

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|-- he_latitude_deg: double (nullable = true)
|-- he_longitude_deg: double (nullable = true)
|-- he_elevation_ft: integer (nullable = true)
|-- he_heading_degT: double (nullable = true)
|-- he_displaced_threshold_ft: integer (nullable = true)

11:48 AM (3s) 7

print("Airport Row Count:", df_airports.count())

print("Runway Row Count:", df_runways.count())
> See performance (2)

Airport Row Count: 85324
Runway Row Count: 47884

11:51 AM (<1s) 8

df_combined = df_airports.join(
df_runways,
df_airports.id == df_runways.airport_id,
"inner"
)

> df_combined: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 37 more fields]

11:52 AM (<1s) 9

Search

ENG IN

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- Learn
- Workspace
- Recents
- Catalog
- Jobs & Pipelines
- Compute
- Discover Beta
- Marketplace
- SQL
- SQL Editor
- Queries
- Dashboards
- Genie Spaces
- Alerts
- Query History
- SQL Warehouses
- Data Engineering
- Runs
- Data Ingestion

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1151 AM (<1s) 8

```
    "inner"
  )
  > df_combined: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 37 more fields]
```

1152 AM (<1s) 9

```
df_combined = df_combined.withColumn(
  "runway_category",
  when(col("length_ft") >= 10000, "Long")
  .when((col("length_ft") >= 5000) & (col("length_ft") <= 9999), "Medium")
  .otherwise("Short")
)
  > df_combined: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 38 more fields]
```

1152 AM (<1s) 10

```
df_combined.createOrReplaceTempView("airports_view")
  > See performance (1)
```

1153 AM (2s) 11

```
spark.sql("""
SELECT
  iso_country,
```

Hot weather Now

Search

ENG IN 12:04 09-05-2026

The screenshot displays the Databricks workspace interface. The top navigation bar includes the Databricks logo, a search bar, and a 'workspace' dropdown. The left sidebar contains a navigation menu with options like 'New', 'Home', 'Learn', 'Workspace' (selected), 'Recents', 'Catalog', 'Jobs & Pipelines', 'Compute', 'Discover Beta', 'Marketplace', 'SQL', 'SQL Editor', 'Queries', 'Dashboards', 'Genie Spaces', 'Alerts', 'Query History', 'SQL Warehouses', 'Data Engineering', 'Runs', and 'Data Ingestion'.

The main workspace area shows a 'practical lab' notebook. The top of the notebook has a menu bar with 'File', 'Edit', 'View', 'Run', 'Help', a 'Python' language selector, and a star icon. Below the menu bar, there are buttons for 'Run all', 'Serverless', 'Schedule', and 'Share'. The notebook content displays a SQL query executed at 11:53 AM (2s):

```
spark.sql("""
SELECT
    iso_country,
    COUNT(*) AS runway_count
FROM airports_view
GROUP BY iso_country
ORDER BY runway_count DESC
LIMIT 10
""").show()
```

Below the query, there is a link to 'See performance (1)'. The results of the query are shown in a table format:

iso_country	runway_count
US	26450
BR	5825
AU	2103
CA	1570
AR	761
FR	659
GB	600
DE	592
RU	489
ID	414

The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 12:04 on 09-05-2026.

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Crimes - 202

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workspace

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Spaces

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

practical lab

File Edit View Run Help Python

Last edit was 1 minute ago

Run all Serverless Schedule Share

11:54 AM (1s) 12

```
spark.sql("""
SELECT
  continent,
  AVG(length_ft) AS avg_runway_length
FROM airports_view
WHERE length_ft IS NOT NULL
GROUP BY continent
ORDER BY avg_runway_length DESC
""").show()
```

See performance (1)

continent	avg_runway_length
AN	7768.857142857143
AS	6925.646996647675
AF	6787.526448362721
EU	4103.151335311572
OC	3558.0401785714284
SA	2857.0891141239963
NA	2599.724218941758

11:54 AM (1s) 13

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Search

ENG IN

12:04 09-05-2026

Launch AWS

EMR on EC2

Clusters | Clou

MongoDB Atl

practical lab

Big Data Tech

Crimes - 202

Databricks - Y

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New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Spaces

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

Hot weather Now

practical lab

File Edit View Run Help Python

Last edit was 1 minute ago

Run all Serverless Schedule Share

11:54 AM (1s) 13

```
spark.sql("""
SELECT
    name,
    municipality,
    COUNT(*) AS lighted_runways
FROM airports_view
WHERE iso_country = 'IN'
AND lighted = 1
GROUP BY name, municipality
""").show()
```

[See performance \(t\)](#)

name municipality lighted_runways
Maharshi Valmiki ... Faizabad 1
Navi Mumbai Inter... Navi Mumbai 2
Dholera Internati... Dholera 1
Mehsana Airport Mehsana 1
Dhulia Airport NULL 1
Raigarh Airport (... NULL 1
Khajuraho Airport Khajuraho 1
Dimapur Airport Dimapur 1
Purnea Airport NULL 1
Hindon Airport / ... Ghaziabad 1
Jodhpur Airport Jodhpur 1
Jammu Airport Jammu 1
Shravasti Airport Shravasti 1
Hashimara Air For... Jalpaiguri 1

ENG IN

12:04 09-05-2026

The screenshot displays the Databricks workspace interface. On the left is a sidebar with navigation options: Home, Learn, Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, and SQL Warehouses. Below these are sections for Data Engineering, Runs, and Data Ingestion.

The main area shows a notebook titled 'practical lab'. The code editor contains a SQL query:

```
spark.sql("""
SELECT
  name,
  iso_country,
  MAX(length_ft) AS max_runway_length
FROM airports_view
GROUP BY name, iso_country
ORDER BY max_runway_length DESC
LIMIT 5
""").show()
```

Below the code editor, the execution results are displayed as a table:

name	iso_country	max_runway_length
Gunflint Seaplane...	US	30000
Libby Camps Seapl...	US	26000
Brookville Reserv...	US	25000
Long Lake Seaplan...	US	25000
Conchas Lake Seap...	US	21120

The bottom status bar indicates the execution time as 11:58 AM (3s) and the cell number as 15: Cell 15.

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Spaces

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

practical lab

File Edit View Run Help Python Last edit was 1 minute ago

[Libby Camps Seaplan...]	US	26000
[Brookville Reserv...]	US	25000
[Long Lake Seaplan...]	US	25000
[Conchas Lake Seap...]	US	21120

11:58 AM (3s) 15: Cell 15

```
df_combined_clean = df_combined.drop(df_runways.id)

df_combined_clean.write.mode("overwrite").parquet(
    "/Volumes/practical/schema1/practicalexam/combined_parquet"
)
```

> See performance (1)

> df_combined_clean: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 37 more fields]

11:58 AM (1s) 16

```
df_parquet = spark.read.parquet(
    "/Volumes/practical/schema1/practicalexam/combined_parquet"
)
```

df_parquet: pyspark.sql.connect.dataframe.DataFrame

```
id: integer
ident: string
type: string
name: string
latitude_deg: double
```

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```
dt_combined_clean: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 37 more fields]
```

11:58 AM (1s) 16 Python

```
df_parquet = spark.read.parquet(
    "/Volumes/practical/schema1/practicalexam/combined_parquet"
)
```

df_parquet: pyspark.sql.connect.dataframe.DataFrame

```
id: integer
ident: string
type: string
name: string
latitude_deg: double
longitude_deg: double
elevation_ft: integer
continent: string
iso_country: string
iso_region: string
municipality: string
scheduled_service: string
icao_code: string
iata_code: string
gps_code: string
local_code: string
home_link: string
wikipedia_link: string
keywords: string
airport_ref: integer
airport_ident: string
length_ft: integer
```

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ENG IN 12:05 09-05-2026

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Spaces

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

practical lab

File Edit View Run Help Python Last edit was 1 minute ago

Run all Serverless Schedule Share

3 minutes ago (1s)

display(df_parquet.limit(10))

See performance (1)

Table

	id	ident	type	name	1.2 latitude_deg	1.2 longitude_deg	elevation_ft	continent	ix
2	343722	JP-1898	heliport	Sakae Heliport	35.85252	140.23244	33	AS	JP
3	350140	JP-2684	heliport	Daikata Heliport	35.77727	140.27391	16	AS	JP
4	508549	JP-3163	small_airport	Kozaki Airfield	35.90741	140.40668	10	AS	JP
5	521063	JP-3393	heliport	Fukushima Heliport	37.708503	140.378959	670	AS	JP
6	547277	JP-3414	heliport	Maebashi Red Cross Hospital Helipad	36.359202	139.096423	327	AS	JP
7	5674	JP-3590	closed	Ishigaki Airport	24.3445	124.187	93	AS	JP
8	324934	JP-3592	closed	Teshikaga Airport	43.479495	144.432911	360	AS	JP
9	313629	JP-3600	small_airport	Satsuma Iojima Airport	30.784722	130.270556	338	AS	JP
10	18255	JPN	heliport	Pentagon Army Heliport	38.874908	-77.056682	40	NA	US

10 rows | 0.60s runtime

Refreshed 3 minutes ago

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